5.13.2 Constructing a parallelogram when its two adjacent sides and one diagonal are given

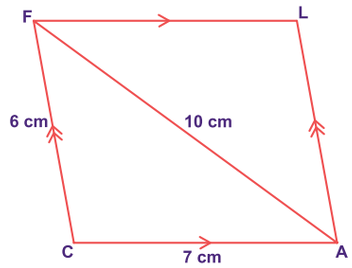
Example 5.31

Construct a parallelogram CALF with CA=7*cm*, CF=6*cm* and AF=10*cm*. Also find its area.

Solution:

Given:

CA=7*cm*, CF=6*cm* and AF=10*cm*



Rough diagram

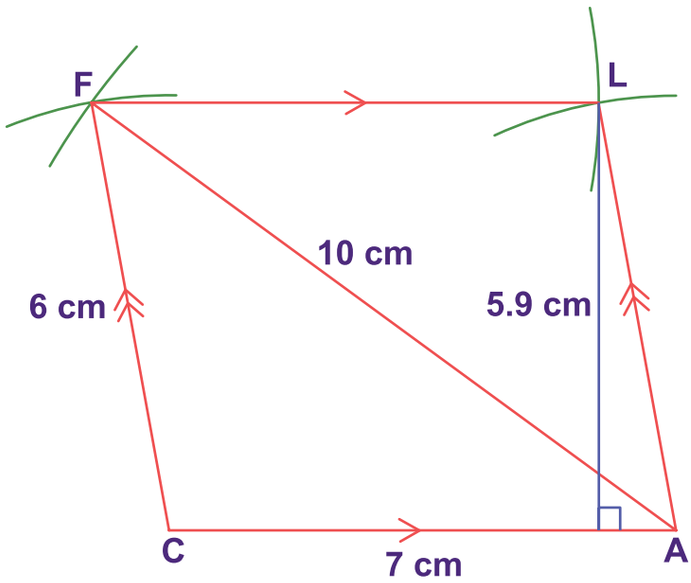


Fig. 5.50

Steps:

1. Draw a line segment CA=7*cm*.
2. With C and A as centres, draw arcs of radii 6*cm* and 10*cm* respectively. Let them cut at F.
3. Join CF and AF.
4. With A and F as centres, draw arcs of radii 6*cm* and 7*cm* respectively. Let them cut at L.
5. Join AL and FL.
6. CALF is the required parallelogram.

Calculation of Area:

Area of the parallelogram CALF = bh sq.units

$= 7 \times 5.9 = 41.3\, sq.cm$

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